Data Augmentation for Image Datasets

Data augmentation is a technique used in machine learning to artificially increase the size and diversity of a dataset by applying transformations to existing data.

For example, in **image classification,** data augmentation involves making small changes to images—such as rotating, flipping, zooming, or changing brightness—so that the model learns to recognize objects in different conditions. This helps improve model accuracy and reduces overfitting (when a model memorizes instead of generalizing).

In simple terms, it's like teaching a model to recognize a cat **whether it's upside down, zoomed in, or in different lighting**. 🐱📸

Lab 01: Data Augmentation for Image Datasets

Objective

1. Understand the concept of data augmentation and its importance in machine learning.
2. Apply data augmentation techniques to a cat vs. dog dataset.
3. Train a simple model with and without data augmentation to observe the impact.

Prerequisites

1. Basic knowledge of Python and deep learning frameworks (e.g., TensorFlow/Keras or PyTorch).
2. A dataset of cat and dog images (e.g., the Kaggle "Dogs vs. Cats" dataset).
3. Libraries: TensorFlow/Keras or PyTorch, Matplotlib, NumPy.

Lab Setup

1. Dataset: Download the [Dogs vs. Cats dataset] from Kaggle.
2. Environment: Set up a Python environment with the required libraries:
   - `pip install tensorflow matplotlib numpy`

Lab Tasks

Task 1: Load and Explore the Dataset

- Load the dataset and visualize a few images.
- Split the dataset into training and validation sets.
- code_for_task_1.py

Exercise:

- Modify the code to display 4 random images from the dataset.

Task 2: Apply Data Augmentation

- Use \``ImageDataGenerator`\` (TensorFlow/Keras) or \``torchvision.transforms`\` (PyTorch) to apply data augmentation.
- Augmentation techniques: rotation, flipping, zooming, shifting, etc.
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